

A PROJECT TITLE

ON

Wine Quality Prediction Using Ridge, Lasso, and Elastic Net Regression

A project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF CSE

Name of Title: Wine Quality Prediction Using Ridge, Lasso, and Elastic Net Regression

1. ABSTRACT:

Machine Learning techniques have become increasingly important in predicting product quality using data-driven approaches. This project focuses on predicting the quality of wine by analyzing its physicochemical properties such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, sulphates, density, pH, and alcohol content. The objective is to build an accurate regression model that estimates wine quality while minimizing prediction error. The dataset is first preprocessed by handling missing values, performing feature scaling using StandardScaler, and separating training and testing data. Three regularized linear regression models, namely Ridge Regression (L2 Regularization), Lasso Regression (L1 Regularization), and Elastic Net Regression (Combination of L1 and L2), are implemented to reduce overfitting and improve model generalization. The models are trained using appropriate regularization parameters and their performance is compared. The effectiveness of each model is evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score. The project also analyzes the impact of regularization on feature selection, coefficient shrinkage, and prediction accuracy. By comparing Ridge, Lasso, and Elastic Net, the proposed system identifies the most suitable regression technique for predicting wine quality. This project demonstrates the practical application of regularized machine learning models for reliable quality prediction and supports data-driven decision making in the food and beverage industry.

2. INTRODUCTION:

Wine quality prediction is an important application of Machine Learning that uses data-driven techniques to estimate the quality of wine based on its physicochemical properties. Traditionally, wine quality is evaluated by expert tasters through sensory analysis, which can be subjective, time-consuming, and expensive. Machine Learning provides a more efficient and consistent approach by learning patterns from historical data and predicting wine quality automatically. This helps reduce human effort while ensuring reliable and accurate quality assessment. The quality of wine depends on several chemical characteristics, including fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol content. These features significantly influence the taste, aroma, texture, and overall quality of the wine. By analyzing the relationship between these physicochemical properties and the quality score, Machine Learning models can accurately estimate the quality of new wine samples.

In this project, three regularized regression algorithms—Ridge Regression, Lasso Regression, and Elastic Net Regression—are implemented to build predictive models. These algorithms are selected because they effectively handle multicollinearity among features, reduce overfitting, and improve the model's ability to generalize to unseen data. Before training the models, the dataset is preprocessed by handling missing values

(if any), splitting the data into training and testing sets, and applying feature scaling to ensure that all input features contribute equally to the prediction process. The performance of each regression model is evaluated using standard regression metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2 Score). These metrics help measure prediction accuracy, error rate, and the model's ability to explain the variation in wine quality. By comparing the results of Ridge, Lasso, and Elastic Net Regression, the project identifies the most effective algorithm for predicting wine quality.

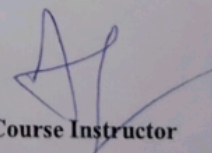
Overall, this project demonstrates how Machine Learning can improve quality assessment in the wine industry by providing a fast, accurate, and cost-effective prediction system. The proposed approach can assist wine manufacturers, quality control laboratories, and researchers in maintaining product standards, optimizing production processes, and making data-driven decisions, ultimately leading to better product consistency and customer satisfaction.

3. SOFTWARE REQUIREMENTS:

Components	Requirements
Programming Language	Python 3.x
Development Environment	Jupyter Notebook, Visual Studio Code (VS Code), or Google Colab
Data Visualization Libraries	Matplotlib, Seaborn
Storage Space	Minimum 500 MB free disk space
Input	Physicochemical properties of wine (fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, alcohol)
Output	Predicted wine quality score
Evolution Metrics	Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score
Operating Systems	Windows 10 or Windows 11



Signature of the Guide



Course Instructor